# EXPERIMENT - 15

**AIM :**

To implement a Decision Tree classifier using Python to predict whether a student passes or fails. write a Python program to implement the A* search algorithm to find the least-cost path from a starting node to a goal node using path cost and heuristic estimation..

**ALGORITHM:**

1. Start the program.
2. Import the Decision Tree classifier.
3. Create a dataset containing study hours and attendance.
4. Assign pass or fail labels to the data.
5. Train the Decision Tree model.
6. Predict the result for a new student.
7. Display the prediction and accuracy.
8. Stop the program.

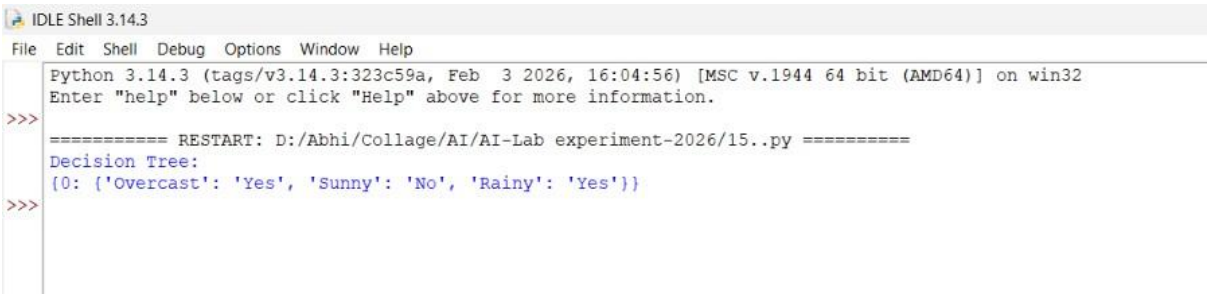
**PROGRAM :**

*untitled (3.14.3)*

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```
import math
from collections import Counter
data = [
    ["Sunny", "No"],
    ["Sunny", "No"],
    ["Overcast", "Yes"],
    ["Rainy", "Yes"],
    ["Rainy", "Yes"],
    ["Rainy", "No"],
    ["Overcast", "Yes"],
    ["Sunny", "No"]
]
def entropy(rows):
    counts = Counter(row[-1] for row in rows)
    total = len(rows)
    result = 0
    for count in counts.values():
        p = count / total
        result -= p * math.log2(p)
    return result
def build_tree(rows, features):
    labels = [row[-1] for row in rows]
    if len(set(labels)) == 1:
        return labels[0]
    if not features:
        return Counter(labels).most_common(1)[0][0]
    best_feature = max(
        features,
        key=lambda f: entropy(rows) -
        sum(
            len([r for r in rows if r[f] == v]) / len(rows)
            * entropy([r for r in rows if r[f] == v])
            for v in set(r[f] for r in rows)
        )
    )
    tree = {best_feature: {}}
    for value in set(row[best_feature] for row in rows):
        subset = [r for r in rows if r[best_feature] == value]
        remaining = [f for f in features if f != best_feature]
        tree[best_feature][value] = build_tree(subset, remaining)
    return tree
features = [0]
tree = build_tree(data, features)
print("Decision Tree:")
print(tree)
```

**OUTPUT:**




**RESULT:**

Thus, the Decision Tree classifier was implemented successfully to predict the student's result based on study hours and attendance.